BIOMÉRIEUX

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VITEK® 2 AST-N439

℞ only IVD

Intended Use

The VITEK® 2 Gram-negative Susceptibility Card is intended for use with the VITEK® 2 Systems in clinical laboratories as an *in vitro* test to determine the susceptibility of clinically significant aerobic Gram-negative bacilli to antimicrobial agents when used as instructed.

Summary and Explanation

Susceptibility testing is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy. Susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used agents. Isolated colonies of each type of organism that may play a pathogenic role are selected from an agar plate and tested for susceptibility. These tests are then examined and the Minimum Inhibitory Concentration (MIC) is determined. The MIC obtained using a dilution test may tell the physician the concentration of an antimicrobial agent needed at the site of infection to inhibit the infecting organism.

MICs have traditionally been determined using antimicrobial concentrations derived from serial twofold dilutions.[2] The MIC is then determined from the lowest concentration that exhibits inhibition of growth. An interpretive criterion (Susceptible, Intermediate, or Resistant) can then be assigned to MIC results to aid in the direction of therapy.

For some antimicrobials (e.g., ESBL) a qualitative result is generated.

The standard and reference procedures are based on susceptibility tests requiring 16 to 24 hours of incubation for bacteria. Various manufacturers have now developed automated procedures designed to generate results more rapidly by using shortened incubation times. Laboratories worldwide use either variations of the standard reference procedure or a commercially available product to determine the MICs of infectious organisms.

AES (Advanced Expert System)

The AES (Advanced Expert System) is a software tool that provides information about the clinical isolate tested. AES determines the consistency level of the AST results, as well as alerts the user to unusual results. AES proposes phenotypes per each class of antimicrobial tested, and applies therapeutic corrections (TCs) based on the proposed phenotypes, and AES parameter set applied.

Since AES is proposing a phenotype based on each class of antimicrobials tested, results can vary depending on the card configuration. It is important to note that a proposal of a phenotype by AES is not considered confirmation of the presence of a particular resistance mechanism. Users are responsible for the results being released from their laboratory and have the ability to stop certain phenotypes for review (refer to VITEK® 2 Systems Software User Manual). AES can provide information about the isolate tested but does not replace the review of results by skilled laboratory personnel.

bioMérieux verifies all changes to the AES Knowledge Base (KB), and biological validation is performed for every AES KB update. Since AES phenotype proposals can vary according to the card configuration, it is recommended that the user perform review of results when updating from one version of software to the next, or when changing to a new card configuration, according to their internal procedures. This review will ensure that AES is providing expected results for their cards, or allow the user to make modifications in AES review settings if deemed appropriate.

Extended-Spectrum Beta Lactamases (ESBLs)

ESBLs are enzymes that arise by mutations in genes for common plasmid-mediated beta-lactamases. Strains of *Klebsiella* spp. and *E. coli* that produce ESBLs may be clinically resistant to therapy with penicillins, cephalosporins, or aztreonam despite apparent *in vitro* susceptibility to some of these agents. Some of these strains will show MICs above the normal susceptible population but below the standard breakpoints for certain extended-spectrum cephalosporins or aztreonam. Before reporting a strain as an ESBL producer, a confirmatory method may be performed. The VITEK® 2 ESBL Test is a

confirmatory test for those ESBLs inhibited by clavulanic acid, and it utilizes cefepime, cefotaxime, and ceftazidime, with and without clavulanic acid, to determine a positive or negative result.

Epidemiological Cut-Off (ECOFF)
An ECOFF is a means of distinguishing between wild-type (WT) and non-wild-type (NWT) isolates using MIC distributions. ECOFFs should not be used as clinical breakpoints. bioMérieux is applying them for surveillance purposes (e.g., separating WT from NWT) when breakpoints are not defined. Several veterinary antimicrobials have no associated breakpoints. Therefore, ECOFFs assist in the determination of wild-type versus non-wild type isolates.

Storage Conditions

Upon receipt, store VITEK® 2 AST cards unopened in their original package liner at 2°C to 8°C.

Principle of the Test

The AST card for VITEK® 2 Systems is an automated test methodology based on the MIC technique reported by MacLowry and Marsh and Gerlach.[15,16] The AST card is essentially a miniaturized and abbreviated version of the doubling dilution technique for MICs determined by the microdilution method.[1]

Each AST card contains a control well containing only microbiological culture medium. The remaining microwells contain premeasured amounts of specific antimicrobials combined with culture medium.

The organism suspension to be tested must be diluted to a standardized concentration in 0.45% saline before being used to rehydrate the antimicrobial medium within the card. The card is then filled, sealed, and placed into the instrument incubator/reader, either automatically (as with VITEK® 2 60 or VITEK® 2 XL) or manually (as with VITEK® 2 Compact). The instrument monitors the growth of each well in the card over a defined period of time (up to 18 hours for Gram-negatives). At the completion of the incubation cycle, MIC values (or test results, as appropriate) are determined for each antimicrobial contained on the card.

Precautions

- For *In Vitro* Diagnostic Use Only.
- For US Only: Caution: US Federal Law restricts this device to sale by or on the order of a licensed practitioner.
- For professional use only.
- Suspensions outside of the appropriate ranges on the VITEK® 2 DENSICHEK®, the VITEK® 2 DENSICHEK® Plus, or the VITEK® DENSICHEK® may compromise card performance.
- The safety and efficacy of antimicrobial drugs, for which antimicrobial susceptibility is tested by this AST device, may or may not have been established in adequate and well-controlled clinical trials for treating clinical infections due to microorganisms outside of those found in the indications and usage in the drug label. The clinical significance of susceptibility information in those instances is unknown. The approved labeling for specific antimicrobial drugs provides the uses for which the antimicrobial drug is approved.
- Do not use the card after the expiration date shown on the package liner.
- Store the card unopened in the package liner. Do not use the card if the protective package liner is damaged or if no desiccant is present.
- Allow the card to come to room temperature before opening the package liner.
- Do not use powdered gloves. Powder may interfere with the optics.
- Use of culture media other than the recommended types must be validated by the customer laboratory for acceptable performance.
- The card performs as intended only when used in conjunction with VITEK® 2 Systems, following the instructions contained in the Instructions for Use.
- It is highly recommended that Good Laboratory Practices (e.g., FDA, CLSI, ISO, etc.) also be followed, according to local guidelines or requirements.
- **Do not use glass test tubes**. Use clear plastic (polystyrene) tubes only. Variation exists among test tubes of standard diameter. Carefully place the tube into the cassette. If resistance is encountered, discard and try another tube that does not require pressure to insert.

- Prior to inoculation, inspect cards for tape tears or damage to the tape and discard any that are suspect. Check the saline level in the tubes after the cassette has been processed to ensure proper filling of card.
  - VITEK® 2 60 or VITEK® 2 XL: Eject improperly filled cards.
  - VITEK® 2 Compact: Do not load improperly filled cards.
- Give special consideration to specimen source and patient therapy regimen. AST cards may contain some antimicrobials that are not proven to be effective for treatment of infections due to all organisms that may be tested. For interpreting and reporting of antimicrobial results that have been shown to be active against organism groups both *in vitro* and in clinical infections, refer to the individual pharmaceutical antimicrobial labeling or local therapy guidelines.
- Interpretation of test results requires the judgment and skill of a person knowledgeable in AST. Additional testing may be required.[17]

Warning: All patient specimens, microbial cultures, and inoculated VITEK® 2 cards, along with associated materials, are potentially infectious and should be treated with universal precautions [18,20]

Reagents

When used with VITEK® 2 instrumentation, the AST card is a complete system for routine susceptibility testing. Each AST card contains selected antimicrobials in varying concentrations, dried with a microbiological culture medium.

Table 1: Contents of the Card

Antimicrobic	Code	Concentration §	Calling Range ≤	Calling Range ≥	FDA Indications for Use
Amikacin	an03n	2, 4, 16, 48	1	64	*Pseudomonas* spp., *E. coli*, *P. mirabilis*, *Klebsiella* spp. , *Enterobacter* spp., *Serratia* spp., *Acinetobacter* spp. (excluding *A. baumannii* complex), *C. freundii*
Amoxicillin/ Clavulanic Acid	amc03n	8/2, 16/2, 64/2	4/2	64/2	**N/A
Cefepime	fep03n	0.25, 1, 4, 16, 32	0.12	32	*Enterobacter* spp., *E. coli*, *Klebsiella* (*Enterobacter*) *aerogenes*, *K. pneumoniae*, *P. mirabilis*, *P. aeruginosa*, *C. koseri*, *C. freundii*, *P. agglomerans*, *K. oxytoca*, *P. vulgaris*, *P. rettgeri*, *P. stuartii*, *S. marcescens*
Ceftazidime	caz02n	0.25, 1, 2, 8, 32	0.12	64	**N/A

Antimicrobic	Code	Concentration §	Calling Range ≤	Calling Range ≥	FDA Indications for Use
Ceftazidime/ Avibactam	cza02n	0.06/4, 0.25/4, 1/4, 4/4, 8/4	0.12	16	*C. freundii*, *E. cloacae*, *E. coli*, *K. oxytoca*, *K. pneumoniae*, *P. mirabilis*, *P. aeruginosa*, *C. koseri*, *Klebsiella* (*Enterobacter*) *aerogenes*, *M. morganii*, *P. stuartii*, *S. marcescens*
Ceftolozane/ Tazobactam	ct01n	0.5/4, 1/4, 4/4, 8/4, 32/4	0.25	32	*E. cloacae*, *E. coli*, *K. oxytoca*, *K. pneumoniae*, *P. mirabilis*, *P. aeruginosa*, *C. freundii*, *C. koseri*, *Klebsiella* (*Enterobacter*) *aerogenes*, *P. vulgaris*, *P. stuartii*, *S. liquefaciens*
Ciprofloxacin	cip02n	0.06, 0.12, 0.5, 1	0.06	4	*C. freundii*, *C. koseri*, *E. cloacae*, *E. coli*, *K. pneumoniae*, *M. morganii*, *P. mirabilis*, *P. vulgaris*, *P. rettgeri*, *P. stuartii*, *P. aeruginosa*, *Salmonella typhi*, *S. marcescens*, *S. sonnei*, *Klebsiella* (*Enterobacter*) *aerogenes*, *K. oxytoca*, *Salmonella enteritidis*
Colistin	cs02n	0.125, 0.5, 2, 8	0.25	8	**N/A
Fosfomycin	fos02n	8, 16, 32	16	256	**N/A
Gentamicin	gm02n	4, 8, 32	1	16	*Citrobacter spp.*, *Enterobacter* spp., *E. coli*, *Klebsiella* spp., *Proteus* spp., *Serratia* spp., *P. aeruginosa*
Imipenem/Relebactam	ipr01n	0.25/4, 1/4, 4/4, 16/4	0.25/4	16/4	*Klebsiella* (*Enterobacter*) *aerogenes*, *E. cloacae*, *E. coli*, *K. pneumoniae*, *P. aeruginosa*, *C. freundii*, *K. oxytoca*, *C. koseri*, *E. asburiae*

Antimicrobic	Code	Concentration §	Calling Range ≤	Calling Range ≥	FDA Indications for Use
Meropenem	mem02n❷	0.5, 2, 6, 12	0.25	16	*E. coli*, *K. pneumoniae*, *P. aeruginosa*, *P. mirabilis*, *Acinetobacter* spp., *C. freundii*, *C. koseri*, *E. cloacae*, *K. oxytoca*, *M. morganii*, *P. vulgaris*, *S. marcescens*, *A. hydrophila*, *C. diversus*, *H. alvei*, *P. multocida*, *Salmonella* spp., *Shigella* spp.
Meropenem/ Vaborbactam	mev01n	0.5/8, 2/8, 8/8, 32/8	0.5	64	*Enterobacter cloacae* Complex, *E. coli*, *K. pneumoniae*, *C. freundii*, *C. koseri*, *Klebsiella* (*Enterobacter*) *aerogenes*, *K. oxytoca*, *M. morganii*, *P. mirabilis*, *Providencia* spp., *S. marcescens*
Piperacillin/ Tazobactam	tzp03n[c]	2/4, 8/4, 24/4, 32/4, 32/8, 48/8	4/4	128/4	*A. baumannii*, *E. coli*, *K. pneumoniaeP. aeruginosa*, *C. koseri*, *M. morganii*, *P. mirabilis*, *P. vulgaris*, , *P. stuartii*, *S. enterica*
Tigecycline	tgc02n	1.5, 4, 8	0.5	8	*C. freundii*, *E. cloacae*, *E. coli*, *K. oxytoca*, *K. pneumoniae*, *C. koseri*, *Klebsiella* (*Enterobacter*) *aerogenes*, *S. marcescens*

Numerical values are expressed in µg/mL.

§ Equivalent standard method concentration by efficacy.

[c] = Category agreement was established at the time of FDA clearance. Essential agreement was not established since test contains less than five discrete dilutions.

❶, ❷ etc. = See performance characteristics identified by the drug code with this symbol.

**N/A = No specific FDA Indications for Use available

Instrument

The VITEK® 2 instruments are a family of *in vitro* diagnostic devices intended to rapidly assess the antimicrobial susceptibility of bacterial and yeast pathogens to available antimicrobial agents. For detailed information on the use and operation of these devices, refer to the appropriate Instrument User Manual.

Specimen Preparation

Table 2: Culture Requirements Table

VITEK® 2 Card	Media	Age of Culture	Incubation Conditions	McFarland Standards	Dilution for AST	Age of Suspension Before Loading Instrument
AST Gram-Negative	TSAB CBA MAC CPS ID	8 to 24 hours	35°C to 37°C aerobic, non-CO_2	0.50 to 0.63	145 µL in 3.0 mL 0.45% saline	VITEK® 2 Compact: ≤ 30 minutes VITEK® 2: ≤ 1 hour
GN and AST GN pair	CBA[1] MAC[1] TSAB CPS ID	18 to 24 hours	35°C to 37°C aerobic, non-CO_2	0.50 to 0.63	145 µL in 3.0 mL 0.45% saline	≤ 30 minutes

[1] These media were used in the identification product database developments and will give optimal performance.

Culture Requirements Table — Media Abbreviations

CBA = Columbia Sheep Blood Agar

MAC = MacConkey Agar

CPS ID = chromID™ CPS (CPS ID Agar)

TSAB = Trypticase Soy Agar with 5% Sheep Blood

Test Procedure

Warning: Failure to properly follow instructions and recommendations provided in this section for performing laboratory tasks may cause erroneous or delayed results.

Materials

Materials provided are:

- VITEK® 2 DENSICHEK® kit, the VITEK® 2 DENSICHEK® Plus Kit, or the VITEK® DENSICHEK® Kit
- VITEK® 2 DENSICHEK® Standards kit, the VITEK® 2 DENSICHEK® Plus Standards Kit, or the VITEK® DENSICHEK® McFarland References Kit
- VITEK® 2 Cassette
- Adjustable volume saline dispenser
- 12 mm x 75 mm clear plastic (polystyrene) disposable test tubes
- VITEK® 2 60 or VITEK® 2 XL only: VITEK® 2 Pipettor/Diluter Accessory Kit (containing instrument pipette tips and saline hookup) and 0.45% saline bag

Materials required, but not provided are:

- Sterile saline (aqueous 0.45% to 0.50% NaCl, pH 4.5 to 7.0)
- Loops, sterile sticks or swabs
- Appropriate agar medium (See the Culture Requirements Table.)
- QC isolates
- VITEK® 2 AST Cards
- Micropipettors to deliver 145 µL

- Disposable pipette tips

Optional Accessories:

- Pre-dispensed saline test tubes (aqueous 0.45% to 0.50% NaCl, pH 4.5 to 7.0)
- Test tube caps
- Vortex

Test Card Setup Procedure

The following procedure contains general information applicable to all susceptibility products. (See the Culture Requirements Table for product-specific information.)

Note: Prepare the inoculum from a pure culture, according to good laboratory practices. In case of mixed cultures, a re-isolation step is required. It is recommended that a purity check plate be done to ensure that a pure culture was used for testing. To enhance and support good laboratory practices, bioMérieux recommends creating a purity plate using the transfer tube/card straw after the card has been filled in the VITEK® 2 system. Please note that underlying growth or other colony types on a purity plate may not be easily visible.

Note: Refer to the user manual for your specific brand of dispensette to ensure the maintenance instructions are followed. The only recommended cleaning procedure for dispensettes is via autoclave. The use of chemicals or cleaning agents (like bleach or soap) can negatively impact the functionality of the dispensette as well as results. bioMérieux recommends autoclaving on a routine basis, at a minimum when a new bottle of saline is started.

Note: To enhance and support good laboratory practices, bioMérieux recommends to check for low-level saline contamination on a routine basis, by dispensing 1 mL of saline into a tubed broth media (for example Tryptic Soy Broth, Brain Heart Infusion, and Thioglycolate) and incubating at 35-37° Centigrade for 2-3 days. Check every day for growth. If the above process is not possible, discard the open bottle of saline and use a new bottle. Autoclaving the dispensette is necessary when starting a new bottle of saline and should be performed on a routine basis. Undetected contamination of the saline can lead to the reporting of inappropriate results.

1. Do one of the following:
   - Select isolated colonies from a primary plate if culture requirements are met.
   - Subculture the organism to be tested to appropriate agar medium and incubate accordingly.
2. Aseptically transfer 3.0 mL of sterile saline (aqueous 0.45% to 0.50% NaCl, pH 4.5 to 7.0) into a clear plastic (polystyrene) test tube (12 mm x 75 mm).
3. Using sterile technique, prepare a homogenous organism suspension with a density equivalent to the appropriate McFarland standard or McFarland Reference using a compatible benchtop densitometer (see the Culture Requirements Table).
   Note: The age of the suspension before loading the instrument for AST testing must be less than one hour when using VITEK® 2 60 or VITEK® 2 XL, and less than 30 minutes when using VITEK® 2 Compact.
4. Choose one of the following:
   - **For an automatic dilution (VITEK® 2 60 or VITEK® 2 XL only):** Place the suspension tube prepared in step 3 into the cassette with or without an identification card. In the next cassette slot, place an empty tube and an AST card. The instrument will automatically dilute the bacterial suspension to prepare an inoculum suitable for the susceptibility card.
   - **For a manual dilution (VITEK® 2 Compact, VITEK® 2 60 or VITEK® 2 XL):** In a second tube containing 3.0 mL of saline, transfer 145 µL of the suspension prepared in step 3. Then place this tube in the cassette with a susceptibility card. The tube with the initial bacterial suspension can also be used for inoculation of an identification card.

   Note: Check the saline level in the tubes after filling. When it is evident by the saline level in the tube that a card has been improperly filled, do not load the card if using VITEK® 2 Compact; **or**, eject the card if using VITEK® 2 60 or VITEK® 2 XL.

   Note: Refer to the appropriate Instrument User Manual for detailed information regarding data entry, processing, etc.
5. Follow your local inspecting agency's guidelines for disposal of hazardous waste.

Quality Control

Quality Control organisms should be processed according to the Test Card Setup Procedure.

Note: If a QC strain appears in the QC Table with no expected results, it is not applicable to use that strain for quality control testing of that antimicrobial.

Table 3: Quality Control

CLSI® Quality Control Organisms VITEK® 2 Results									
Antimicrobic	Code	*E. coli* ATCC® 25922™	*P. aeruginosa* ATCC® 27853™	*E. coli* ATCC® 35218™	*K. pneumoniae* ssp. *pneumoniae* ATCC® 700603™	*K. pneumoniae* ATCC® BAA-1705™	*K. pneumoniae* ATCC® BAA-2814™	*E. coli* NCTC 13846‡	*P. aeruginosa* ATCC® BAA-3144™
Amikacin	an03n	≤ 1 - 4 (FDA/CLSI Broth Microdilution expected QC range = 0.5 - 4 µg/mL)	≤ 1 - 4 (FDA/CLSI Broth Microdilution expected QC range = 1 - 4 µg/mL)	-	-	-	-	-	-
Amoxicillin/ Clavulanic Acid	amc03n	≤ 4/2 - 8/2	-	≤ 4/2 - 32/2	-	-	-	-	-
Cefepime	fep03n	≤ 0.12 (FDA/CLSI Broth Microdilution expected QC range = 0.015 - 0.12 µg/mL)	0.5 - 4	-	-	-	-	-	-
Ceftazidime	caz02n	≤ 0.12 - 0.5	1 - 4	-	-	-	-	-	-
Ceftazidime/ Avibactam	cza02n	≤ 0.12 - 0.5 (FDA/CLSI Broth Microdilution expected QC range = 0.06 - 0.5 µg/mL)	0.5 - 4	≤ 0.12 (FDA/CLSI Broth Microdilution expected QC range = 0.03 - 0.12 µg/mL)	0.25 - 2††	-	-	-	-
Ceftolozane/ Tazobactam	ct01n	≤ 0.25 - 0.5 (FDA/CLSI Broth Microdilution expected QC range = 0.12/4 - 0.5/4 µg/mL)	≤ 0.25 - 1	≤ 0.25 (FDA/CLSI Broth Microdilution expected QC range = 0.06/4 - 0.25/4 µg/mL)	0.5 - 2	-	-	-	-
Ciprofloxacin	cip02n	≤ 0.06 (FDA/CLSI Broth Microdilution expected QC range = 0.004 - 0.015 µg/mL)	0.25 - 1	-	-	-	-	-	-
Colistin	cs02n	≤ 0.25 - 2	0.5 - 4	-	-	-	-	2 - ≥ 8	-
Fosfomycin	fos02n	≤ 16	-	-	-	-	-	-	-
Gentamicin	gm02n	≤ 1 (FDA/CLSI Broth Microdilution expected QC range = 0.25 - 1 µg/mL)	≤ 1 - 2 (FDA/CLSI Broth Microdilution expected QC range = 0.5 - 2 µg/mL)	-	-	-	-	-	-
Imipenem/ Relebactam	ipr01n	-	-	-	-	≤ 0.25/4$^{\Delta\infty}$ (FDA/CLSI Broth Microdilution expected QC range = 0.03/4 – 0.25/4 µg/mL)	≤ 0.25/4 - 0.5/4$^{\Delta}$ (FDA/CLSI Broth Microdilution expected QC range = 0.06/4 – 0.5/4 µg/mL)	-	2/4 - 8/4
Meropenem	mem02n	≤ 0.25	≤ 0.25 - 1	-	-	-	-	-	-

CLSI® Quality Control Organisms VITEK® 2 Results									
Antimicrobic	Code	*E. coli* ATCC® 25922™	*P. aeruginosa* ATCC® 27853™	*E. coli* ATCC® 35218™	*K. pneumoniae* ssp. *pneumoniae* ATCC® 700603™	*K. pneumoniae* ATCC® BAA-1705™	*K. pneumoniae* ATCC® BAA-2814™	*E. coli* NCTC 13846‡	*P. aeruginosa* ATCC® BAA-3144™
Meropenem/ Vaborbactam	mev01n	≤ 0.5/8◊ (FDA/CLSI Broth Microdilution expected QC range = 0.008 – 0.06 µg/mL)	≤ 0.5/8 - 1/8◊ (FDA/CLSI Broth Microdilution expected QC range = 0.12 – 1 µg/mL)	≤ 0.5/8◊ ††† FDA/CLSI Broth Microdilution expected QC range = 0.008 – 0.06 µg/mL)	≤ 0.5/8◊ (FDA/CLSI Broth Microdilution expected QC range = 0.015 – 0.06 µg/mL)	≤ 0.5/8 ††† (FDA/CLSI Broth Microdilution expected QC range = 0.008 – 0.06 µg/mL)	≤ 0.5/8◊ ††† (FDA/CLSI Broth Microdilution expected QC range = 0.12 – 0.5 µg/mL)	-	-
Piperacillin/ Tazobactam	tzp03n	≤ 4/4[1]	≤ 4/4 - 8/4[2]	≤ 4/4[3]	-	-	-	-	-
Tigecycline	tgc02n	≤ 0.5 (FDA/CLSI Broth Microdilution expected QC range = 0.03 - 0.25 µg/mL)	-	-	-	-	-	-	-

Numerical values are expressed in µg/mL.

[1]FDA/CLSI Broth Microdilution expected QC range = 1/4 - 4/4 µg/mL

[2]FDA/CLSI Broth Microdilution expected QC range = 1/4 - 8/4 µg/mL

[3]FDA/CLSI Broth Microdilution expected QC range = 0.5/4 - 2/4 µg/mL

†† Per CLSI M100, *K. pneumoniae* ATCC 700603 should be tested against Ceftazidime/Avibactam and Ceftazidime alone to confirm the activity of Avibactam in the combination and to ensure that the plasmid encoding the beta-lactamase has not been lost in this strain. The acceptable range for Ceftazidime alone is greater than 16 µ g/mL.

††† Refer to current CLSI M100 Table, *Nonfastidious QC for β-Lactam Combination Agents* for recommendation to ensure *E. coli* ATCC 35218, *K. pneumoniae* ATCC BAA-1705, and/or *K. pneumoniae* ATCC BAA-2814 have not lost the (or their) respective β-lactamase resistance plasmid before conducting QC testing.

◊Meropenem/Vaborbactam does not include the full CLSI/FDA-recommended QC dilution ranges; include this organism for QC testing.

ΔDoes not include the full CLSI/FDA recommended dilution range for QC testing with this organism.

‡NCTC = National Collection of Type Cultures, Public Health England

Certification Statement

This is to certify that bioMérieux complies with ISO 13485 and FDA Quality System Regulation (QSR) requirements for design, development, and manufacture of antimicrobial susceptibility systems.

Frequency of QC Testing

Refer to *Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically,* CLSI® and/or your local guidelines.[2]

Preparation of QC Organisms

1. Rehydrate the organism according to the manufacturer's instructions.
2. Subculture to Trypticase Soy Agar with 5% sheep blood (TSAB).
3. Incubate at 35°C for 24 hours.
4. Check for purity.
5. Subculture to a TSAB plate.
6. Incubate at 35°C for 8-24 hours for Gram-negatives.

Short-Term Storage Conditions

1. Streak to a TSAB plate or slant.
2. Incubate for 24 hours.
3. Refrigerate at 2°C to 8°C for up to two weeks.

4. Subculture once as described above and use for QC.

Long-Term Storage Conditions

1. Make a heavy suspension in Tryptic Soy Broth (TSB) with 15% glycerol.
2. Freeze at -70°C.
3. Subculture to TSAB twice before running QC.
   Note: Avoid repeated thawing and refreezing by either freezing in single-use aliquots or removing a small portion of frozen organism preparation with a sterile applicator stick.

Results

Susceptibility Analytical Techniques

The system evaluates each organism's growth pattern in the presence of the antimicrobial in relation to the growth control well. Several parameters based on the growth characteristics are used to determine the MIC or qualitative result (for example, ESBL POS/NEG). The MIC result must be linked to an organism identification to determine a category interpretation. Accurate identification is critical, especially with certain organism/antimicrobial combinations.

In cases where the identification of an organism is in question, confirmatory testing is necessary to ensure correct interpretation of susceptibility results.

A category interpretation will be reported along with a MIC, according to the interpretations defined by the Food and Drug Administration (FDA), CLSI®, Comité de l'Antibiogramme de la Société Française de Microbiologie (CASFM), European Committee for Antimicrobial Susceptibility Testing (EUCAST), or to an adaptation of the global settings according to other local guidelines.

Note: As some category interpretation definitions differ between US FDA, CLSI, and EUCAST, refer to the respective publications, websites, and/or the VITEK® 2 Software User Manual (Chapter: Maintaining the Workstation), for more detailed information.

Note: When FDA and CLSI® breakpoints differ, VITEK® 2 Systems AST tests are cleared for use with FDA breakpoints applied.

Combination Antimicrobials

The MICs for the combination antimicrobials are listed on the laboratory and patient reports as the first concentration (e.g., ampicillin/sulbactam ≤ 8/4 µg/mL is reported as ≤ 8 µg/mL.). The actual concentrations for each value in the antimicrobial calling range are as follows:

- amoxicillin/clavulanic acid (amc03n) (µg/mL) 4/2, 8/2, 16/2, 32/2, 64/2
- piperacillin/tazobactam (µg/mL) 4/4, 8/4, 16/4, 32/4, 64/4, 128/4
- ceftazidime/avibactam (µg/mL) 0.12/4, 0.25/4, 0.5/4, 1/4, 2/4, 4/4, 8/4, 16/4, 32/4
- ceftolozane/tazobactam (µg/mL) 0.25/4, 0.5/4, 1/4, 2/4, 4/4, 8/4, 16/4, 32/4
- imipenem/relebactam (µg/mL) 0.25/4, 0.5/4, 1/4, 2/4, 4/4, 8/4, 16/4
- meropenem/vaborbactam (µg/mL) 0.5/8, 1/8, 2/8, 4/8, 8/8, 16/8, 32/8, 64/8

Antimicrobial Deduction

Antimicrobials that have been deduced will only report an interpretive result and will be noted with a **+**.

Clinical Efficacy and Indications for Use

AST cards may contain some antimicrobials that are not proven to be effective for treatment of infections caused by all organisms that may be tested. For interpreting and reporting of antimicrobial results that have shown to be active against organism groups both *in vitro* and in clinical infections, refer to the individual pharmaceutical antimicrobial labeling or the local therapy guidelines.

Indications for use specific to the FDA are found in each VITEK® 2 AST card package insert in the column labeled, "FDA Indications for Use". This list contains the antimicrobial/organism combinations that have been cleared by the FDA for testing and reporting on the VITEK® 2 System. FDA clearance has been granted according to the FDA approved pharmaceutical labeling and the VITEK® 2 AST clinical trial data. To report only organisms listed in the FDA Indications for Use section of the package insert, enable bioART Indications for Use suppression rules.

Urinary Use Only Antimicrobials

Certain antimicrobial agents are limited to use in treating urinary tract infections. Accordingly, these agents should not be reported against pathogens recovered from infection sites other than the urinary tract (except as noted). Please refer to the current *CLSI Performance Standards for Susceptibility Testing, M100* and/or local guidelines for additional information.[3]

For urine only, per CLSI®:[3]

- *Enterobacteriaceae*: lomefloxacin or ofloxacin, norfloxacin, nitrofurantoin, sulfisoxazole, trimethoprim, cefazolin
- *Pseudomonas aeruginosa*: lomefloxacin or ofloxacin, norfloxacin
- Other non-*Enterobacteriaceae*: lomefloxacin or ofloxacin, norfloxacin, sulfisoxazole, tetracycline

Limitations

A VITEK® 2 AST card cannot be used with a direct clinical specimen or sample or other sources containing mixed flora. Any change or modification in the procedure may affect the results.

A result for an antibiotic/organism combination, which may have a limitation, may be suppressed from reporting. This can be accomplished through the use of bioART rules in the VITEK® 2 Systems software. Refer to the software user manual for instructions.

Perform an alternative method of testing prior to reporting of results for the following antibiotic/organism combination(s):

- Amikacin (an03n): *Acinetobacter baumannii* Complex, *Proteus vulgaris*, *Providencia* spp.
- Cefepime (fep03n): *Bordetella bronchiseptica*, *Hafnia alvei*, *Morganella* spp.
- Ceftolozane/Tazobactam (ct01n): *Morganella morganii*, *Providencia rettgeri*, *Serratia marcescens*
- Ceftazidime/Avibactam (cza02n): *Providencia rettgeri*
- Colistin (cs02n): *Pseudomonas aeruginosa* when the MIC is ≤ 2 µg/mL
- Imipenem/Relebactam (ipr01n): *Proteus spp., Providencia spp., Morganella spp., Serratia spp.*
- Piperacillin/Tazobactam (tzp03n): *Serratia marcescens*, *Pseudomonas aeruginosa* (when the VITEK® 2 MIC is 32 µg/mL or 64 µg/mL and CLSI breakpoints of ≤ 16S , 32-64 I, ≥ 128 R are applied)

Perform an alternative method of testing prior to reporting results when a resistant result is obtained with the following antibiotic/organism combination(s):

- Meropenem (mem02n): *Aeromonas* spp.
- Piperacillin/Tazobactam (tzp03n): *Pseudomonas aeruginosa* when applying CLSI breakpoints ≤ 64 S, ≥ 128 R

The ability of the AST card to detect resistance with the following combination(s) is unknown because resistant strains were not available at the time of comparative testing:

- Ciprofloxacin (cip02n): *Enterobacter cloacae*, *Providencia rettgeri*, *Salmonella enteritidis*, *S. typhi*, *Shigella sonnei*
- Tigecycline (tgc02n): *C. freundii*, *E. cloacae*, *E. coli*, *K. oxytoca*, *K. pneumoniae*, *C. koseri*, *K. (Enterobacter) aerogenes*, *S. marcescens*

The ability of the AST card to detect resistance with the following combination(s) is unknown because an insufficient number of resistant strains were available at the time of comparative testing:

- Imipenem/Relebactam (ipr01n): *C. freundii, C. koseri, E. asburiae*

The ability of the AST card to detect resistance with the following combination(s) is unknown because an adequate number of resistant strains was not available at the time of comparative testing:

- Ceftazidime (caz02n): *Aeromonas* spp. (SW ≥ V2S 9.03)
- Cefepime (fep03n): *Aeromonas* spp., *Vibrio* spp. (SW ≥ V2S 9.03)
- Meropenem (mem02n): *Aeromonas* spp., *Vibrio* spp. (SW ≥ V2S 9.03)

The ability of the AST card to detect resistance with the following combination(s) is unknown because an insufficient number of resistant strains were available at the time of comparative testing:

If such a strain is observed, it should be submitted to a reference laboratory for further testing.

- Ceftolozane/Tazobactam (ct01n): *Citrobacter koseri*, *Proteus mirabilis*, *Proteus vulgaris*, *Serratia liquefacians*
- Gentamicin (gm02n): *Citrobacter koseri*, *K. (Enterobacter) aerogenes*, *Enterobacter cloacae*, *Serratia marcescens*

The ability of the AST card to detect resistance with the following combination(s) is unknown because resistant strains were not available at the time of comparative testing:

If such a strain is observed, it should be submitted to a reference laboratory for further testing.

- Gentamicin (gm02n): *Proteus vulgaris*

Additional limitations for Cefepime (fep03n) are as follows:

Although within Essential Agreement, the lack of an intermediate category has shown major and very major discrepancies when compared to the reference method. Testing should be repeated using an alternative testing/reference method prior to reporting results for the following antibiotic/organism combination(s):

- Cefepime (fep03n): *Pseudomonas aeruginosa* when the VITEK® 2 MIC is 8 µg/mL or 16 µg/mL

The ability of the AST card to detect resistance with the following combination(s) is unknown because resistant strains were either not available or an insufficient number were encountered at the time of comparative testing:

- Amikacin (an03n): *Pseudomonas* spp., *E. coli*, *Proteus* spp. (including *P. mirabilis*), *Klebsiella* spp., *Enterobacter* spp., *K. (Enterobacter) aerogenes, Serratia* spp., *C. freundii*, *Providencia* spp.
- Meropenem/Vaborbactam (mev01n): *E. cloacae* complex, *C. freundii*, *C. koseri*, *K. (Enterobacter) aerogenes*, *K. oxytoca*, *M. morganii*, *P. mirabilis*, *Providencia spp*., and *S. marcescens*.

Although within Essential Agreement when compared to the reference method, errors (major and/or very major) were observed due to the lack of intermediate category. Testing should be repeated using an alternative testing/reference method prior to reporting results for the following antibiotic/organism(s):

- Ceftazidime/Avibactam (cza02n): *Pseudomonas aeruginosa* when the VITEK® 2 MIC is ≥ 16µg/mL due to observed major errors (4.1%), with an adjusted categorical major error rate of 1.2%

EUCAST Limitations

It is recommended that the existing bioART suppression rules be enabled, or new rules created and enabled, for these limitations if the EUCAST breakpoints are applied.

Perform an alternative method of testing prior to reporting of results for the following antibiotic/organism combination(s):

- Amikacin (an03n): *Morganellaceae* (*Proteus* spp., *Providencia* spp., *Morganella* spp.)
- Amoxicillin/Clavulanic Acid (amc03n): *Citrobacter freundii, Serratia marcescens*
- Ceftazidime (caz02n): *Aeromonas* spp. (SW ≥ V2S 9.03)
- Ceftazidime/Avibactam (cza02n): *Providencia rettgeri*
- Ceftolozane/Tazobactam (ct01n): *Providencia rettgeri*
- Colistin (cs02n): *Pseudomonas aeruginosa* when the MIC is ≤ 2 µg/mL
- Imipenem/Relebactam (ipr01n): *Proteus* spp., *Providencia* spp., *Serratia* spp., *Morganella* spp. (SW V2S 9.02 and 9.03)
- Tigecycline (tgc02n): *Proteus vulgaris*, *Providencia stuartii*

The ability of the AST card to detect resistance with the following combination(s) is unknown because resistant strains were either not available or an insufficient number were encountered at the time of comparative testing:

- Amikacin (an03n): *Pseudomonas* spp., *E. coli*, *Klebsiella* spp., *Enterobacter* spp., *Serratia* spp., *C. freundii*

The ability of the AST card to detect resistance with the following combination(s) is unknown because an adequate number of resistant strains was not available at the time of comparative testing:

- Cefepime (fep03n): *Aeromonas* spp. (SW ≥ V2S 9.03)
- Ciprofloxacin (cip02n): *Aeromonas* spp. (SW ≥ V2S 9.03)

Perform an alternative method of testing prior to reporting results when a resistant result is obtained with the following antibiotic/organism combination(s):

- Colistin (cs02n): *Acinetobacter junii*

Expected Values

Expected results for susceptibility tests will vary based on location and institution. VITEK® 2 Systems were tested at several geographically diverse locations to ensure that trends that occurred by location were integrated into the performance characteristics of the system. Organism resistance patterns will differ by institution; therefore, expected values will be directly related to the population of organisms at each site.

Performance Characteristics

The performance characteristics of the antimicrobial agents included in VITEK® 2 AST cards were established using the manual and autodilution modes (on a VITEK® 2 System) at multiple clinical laboratories. The VITEK® 2 AST card results were compared to results from a CLSI® reference method. Essential agreement (EA) represents VITEK® 2 results which agree exactly or are within a ± twofold dilution (± two doubling dilutions for antifungal) of the reference result.

Category agreement (CA) occurs when the VITEK® 2 and the reference interpretative results agree (Susceptible, Intermediate, and Resistant). There are instances when the category agreement for an antimicrobial falls below the essential agreement. This can occur when a significant number of MICs cluster around a category breakpoint during clinical trial testing, resulting in interpretative errors. For a description of interpretive errors, refer to the footnotes below the table that follows (Performance Characteristics). When the majority of the errors are of the minor type, a high corresponding essential agreement percentage demonstrates that the antimicrobial retains an acceptable overall performance.

There are instances when the performance is based solely on category agreement (CA) because at the time performance was established, less than five discrete twofold dilutions were evaluated. A minimum of five dilutions is necessary to calculate essential agreement (EA) based on ± one twofold dilution. These instances are designated by a "c" footnote in the Contents of the Card table. The following performance tables have values for CA only when EA was not established at the time of FDA clearance.

The reproducibility of the VITEK® 2 system was established by testing a set of on-scale organisms.*

*Data on file at bioMérieux, Inc

Table 4: Performance Characteristics for Gram-Negative Antimicrobial Susceptibility Testing

Antimicrobial	Antimicrobial Code	Antibiotic Version	Bp[1]	Comment[2]	Essential Agreement				Category Agreement				% Reproducibility
					% Error				% Error				
					% EA	VME	ME	mE	% CA	VME	ME	mE	
Amikacin	AN	an03n	CLSI	E	91.9	1.5	0.3	1.5	95.8	1.5	0.3	3.8	100
			FDA*	#, E	94.9	N/A	N/A	N/A	98.4	0.0	0.4	1.2	100
			*VITEK® 2 Amikacin MIC values for evaluable *C. freundii*, *E. cloacae* complex, *E. coli*, *K. pneumoniae*, *P. aeruginosa* and *S. marcescens* tended to be in exact agreement or at least one doubling dilution higher compared to the CLSI reference broth microdilution.								
Cefepime	FEP	fep03n	CLSI	E	93.2	2.7	0.1	1.4	95.0	2.7	0.1	4.6	100
				#, E	93.8	N/A	N/A	N/A	96.1	4.0	2.7	1.1	
Note: The EA for VITEK® 2 Cefepime (fep03n) with *Pseudomonas aeruginosa* was 88.4%. The trending of on-scale results demonstrated: 41.3% equivalent, 40.6% one doubling dilution higher, and 6.9% one doubling dilution lower when compared to the CLSI reference broth microdilution.													
Ceftazidime	CAZ	caz02n	CLSI	E	93.1	3.2	0.4	0.9	96.6	3.2	0.4	2.8	100

Antimicrobial	Antimicrobial Code	Antibiotic Version	Bp[1]	Comment[2]	Essential Agreement				Category Agreement				% Reproducibility
					% Error				% Error				
					% EA	VME	ME	mE	% CA	VME	ME	mE	
Ceftazidime/Avibactam	CZA	cza02n	Global CLSI/ Global European	E	94.5	0.0	0.5	0.0	98.9	0.0	1.2	0.0	99.6
			CLSI (FDA)	#, E	94.5	N/A	N/A	N/A	98.7	0.0	1.4*	0.0	99.6
			• *The overall categorical major error rate for *Enterobacteriaceae* and *Pseudomonas aeruginosa* combined was 1.4% (14/998). Nine (9) major errors were one dilution apart from the reference method and as such fall within essential agrement. Based on the essential agreement, and the lack of an intermediate breakpoint for Ceftazidime/Avibactam, the adjusted categorical major error rate is 0.5% (5/998) • VITEK® 2AST-GN Ceftazidime/Avibactam MIC values tended to be in exact agreement or at least one doubling dilution higher when testing *Enterobacteriaceae* or *Pseudomonas aeruginosa* compared to the reference broth micro-dilution. • Enzyme group characterization was not available for the following organisms at the time of comparative testing, and therefore the performance of the AST card with Ceftazidime/Avibactam is unknown: *Enterobacteriaceae* (OXA, metallo beta-lactamases); *Pseudomonas aeruginosa* (metallo beta-lactamases). • Ceftazidime/Avibactam is not active against bacteria that produce metallo beta-lactamase enzymes and may not have activity against Gram-negative bacteria that overexpress efflux pumps or have porin mutations.									
Ceftolozane/ Tazobactam	CT	ct01n	CLSI (FDA)	#, E	94.8	N/A	N/A	N/A	98.4	1.4	0.4	1.0	100
			Note(s): • Ceftolozane/Tazobactam is not active against *Enterobacteriaceae* bacteria that produce metallo-beta lactamases. • Enzyme group characterization was not available for the following organisms at the time of comparative testing, and therefore the performance of the AST card with Ceftolozane/Tazobactam is unknown: *Enterobacteriaceae* (OXA). • Enzyme group characterization was not available for the following organisms at the time of comparative testing, and therefore the performance of the AST card with Ceftolozane/Tazobactam is unknown: *Pseudomonas aeruginosa* (up-regulation of MexXY and MexAB). • One *Pseudomonas aeruginosa* isolate tested during the clinical evaluation lacked the OprD porin and contained the OXA-101 enzyme which has been reported as being resistant to Ceftolozane/Tazobactam and resulted in a Very Major Error for the VITEK® 2 AST-GN Ceftolozane/Tazobactam test. • VITEK® 2 Ceftolozane/Tazobactam MIC values tended to be in exact agreement or at least one doubling dilution higher when testing *P. aeruginosa* compared to the reference broth micro-dilution.									
			CLSI	E	96.7	0.0	0.0	0.3	99.0	0.0	0.0	1.0	100
Ciprofloxacin	CIP	cip02n	CLSI	E	96.6	0.4	0.1	0.1	97.3	0.4	0.1	2.5	100
			CLSI (FDA)*	#, E	97.1	N/A	N/A	N/A	97.1	0.5	0.2	2.6	100
*VITEK® 2 Systems Ciprofloxacin MIC values for all organisms tended to be one or more doubling dilution higher compared to reference broth microdilution.													
Colistin	CS	cs02n	CLSI/ Global CLSI	E	(732/ 793) 92.3	N/A	N/A	(25/7 93) 3.2	(747/ 793) 94.2	N/A	N/A	(46/7 93) 5.8	100
			CA-SFM	E	(736/ 793) 92.8	(5/10 2) 4.9	(16/6 91) 2.3	N/A	(757/ 793) 95.5	(15/1 02) 14.7	(21/6 91) 3.0	N/A	

Antimicrobial	Antimicrobial Code	Antibiotic Version	Bp[1]	Comment[2]	Essential Agreement				Category Agreement				% Reproducibility
					% Error				% Error				
					% EA	VME	ME	mE	% CA	VME	ME	mE	
Gentamicin	GM	gm02n	CLSI (FDA)	#, E	99.4	N/A	N/A	N/A	98.6	1.3	0.0	1.3	100
			Note(s): • Of 163 *K. pneumoniae* isolates tested, 13 were evaluable for trending analysis. Based on this analysis, some VITEK® 2 Gentamicin MIC values tended to be at least one doubling dilution higher when compared to the reference broth microdilution. • Of 90 *Proteus* spp. isolates tested, 13 were evaluable for trending analysis. Based on this analysis, some VITEK® 2 Gentamicin MIC values tended to be at least one doubling dilution lower when compared to the reference broth microdilution.											
			CLSI	E	97.9	0.6	0.0	0.8	97.1	0.6	0.0	2.8	100
Imipemen/Relebactam	IPR	ipr01n	CLSI (FDA)*	#, E All organism groups combined	95.6 (474/496)	N/A	N/A	N/A	98.0 (486/496)	0.0 (0/61)	0.2 (1/424)	1.8 (9/496)	99.3
				Enterobacteriaceae	95.3 (347/364)	N/A	N/A	N/A	98.9 (360/364)	0 (0/43)	0.3 (1/319)	0.8 (3/364)	
				P. aeruginosa	96.2 (127/132)	N/A	N/A	N/A	95.5 (126/132)	0.0 (0/18)	0.0 (0/105)	4.5 (6/132)	
			Global CLSI	E	95.6 (474/496)	0.0 (0/61)	0.2 (1/424)	0.6 (3/496)	98.0 (486/496)	0.0 (0/61)	0.2 (1/424)	1.8 (9/496)	
			*A trending analysis was conducted using combined data (clinical and challenge) for *Enterobacteriaceae* and *Pseudomonas aeruginosa*. This trending analysis determined the VITEK® 2 Imipenem/Relebactam, when compared to the reference broth microdilution, tended to be at least one doubling dilution lower for *Citrobacter koseri* and *Klebsiella pneumoniae*, and at least one doubling dilution higher for *Enterobacter cloacae*.											
Meropenem	MEM	mem02n	CLSI	#, E ❷	97.6	1.4	0.0	0.3	96.8	1.4	0.0	2.8	99.6
			Global	#, E ❷	97.6	0.0	0.0	0.5	97.7	0.0	0.0	2.3	
Meropenem/ Vaborbactam*	MEV	mev01n	CLSI (FDA)	#, E	(441/449) 98.2	N/A	N/A	N/A	(443/449) 98.7	(0/33) 0.0	(0/413) 0.0	(6/449) 1.3	97.78
			Global CLSI	E	(441/449) 98.2	(0/0) 0.0	(0/0) 0.0	(4/449) 0.9	(443/449) 98.7	(0/33) 0.0	(0/413) 0.0	(6/449) 1.3	
*The following enzyme group characterization was not available for *Enterobacteriaceae* at the time of comparative testing, and therefore the performance of the AST-GN Meropenem/Vaborbactam is unknown: SME, CMY, ACT, and porin mutations combined with overexpression of efflux pumps.													
Piperacillin/ Tazobactam*	TZP	tzp03n	CLSI	#, E	N/A	N/A	N/A	N/A	93.2	0.9	2.2	5.0	96.7
* The overall essential agreement for VITEK® 2 piperacillin/tazobactam was not calculated since the test contains less than five discrete concentrations. However, of the 2453 isolates tested, 348 (14.2%) yielded on-scale VITEK® 2 MIC values (i.e., results of 8, 16, 32, and 64 µg/ml). Those results were evaluated for MIC trends and the following results were obtained when VITEK® 2 was compared to CLSI broth microdilution method; 74.4% (259/348) of the VITEK® MICs were within +/- one doubling dilution; 18.4% (64/348) of the VITEK® MICs were higher by more than one doubling dilution; and 7.2% (25/348) of the VITEK® MICs were lower by more than one doubling dilution.													
Tigecycline	TGC	tgc02n	CLSI	#, E	98.9	N/A	N/A	N/A	98.6	0.0	0.3	1.1	99.6
			Note(s): • Overall, VITEK® 2 Tigecycline MICs showed no trending. However, compared to the reference broth microdilution, *K. pneumoniae* results tended to be one dilution higher; results for *S. marcescens* tended to be one dilution lower.										
			Global	E	97.0	0.0	0.1	0.8	93.3	0.0	0.1	6.6	99.6

[1] Abbreviations — Bp = breakpoint committee; EA = essential agreement; CA = category agreement; VME = Very Major Error (susceptible result with resistant reference result); ME = Major Error (resistant result with susceptible reference result); mE = minor Error (susceptible or resistant result with an intermediate reference result, or an intermediate result with a susceptible or resistant reference result).

[2] Comment — Specific organism groups are designated as P. aeurg. for *Pseudomonas aeruginosa*, others for species other than *Pseudomonas aeruginosa*, and Acineto. for *Acinetobacter*.

Key:

= US Food and Drug Administration 510(k) cleared
CLSI® = Clinical and Laboratory Standards Institute
CA-SFM = Comité de l'Antibiogramme de la Société Française de Microbiologie
E = External performance data
I = Internal performance data
– = Not available
❶ ❷ = Symbol identifies the performance characteristics for a specific antimicrobial version.
Ref. = Reference method for clinical performance study.
N/A = Not applicable
ECOFF = Epidemiological Cut-off value for differentiation of wild type vs. non-wild type

For systemic infections, aminoglycosides (AMGs) should always be used in combination with another active agent or therapy. Systemic breakpoints are based on Epidemiological Cut Off values (ECOFFs), and are intended to distinguish between organisms with or without acquired resistance mechanisms (non wild type and wild type, respectively). For infections originating from the urinary tract, the breakpoints are based on MIC distributions of relevant microorganisms and PK/PD calculations. Calculations assume that the AMGs are being prescribed as monotherapy. For additional information, refer to *EUCAST Aminoglycoside Guidance Document*, 2020.

Table 5: EUCAST Performance Characteristics for Gram-Negative Antimicrobial Susceptibility Testing

Antimicrobial	Antimicrobial Code	Antibiotic Version	Comment[1]	Essential Agreement				Category Agreement			
				% Error				% Error			
				%EA[2]	VME	ME	mE	%CA	VME	ME	mE
Amikacin	AN	an03n	*Acinetobacter* spp., *Enterobacterales*, *Pseudomonas* spp.	(597/633) 94.3	(2/74) 2.7	(2/559) 0.4	N/A	(623/633) 98.4	(2/74) 2.7	(8/559) 1.4	N/A
Amoxicillin/Clavulanic Acid	AMC	amc03n	*Enterobacterales* (UTI)	(503/520) 96.7	(6/148) 4.1	(4/372) 1.1	N/A	(501/520) 96.3	(10/148) 6.8	(9/372) 2.4	N/A
			Enterobacterales (non-UTI)	(505/520) 97.1	(4/212) 1.9	(3/308) 1.0	N/A	(498/520) 95.8	(8/212) 3.8	(14/308) 4.5	N/A
Cefepime	FEP	fep03n	*Aeromonas* spp.*, *Enterobacterales*	(614/642) 95.6	(0/86) 0.0	(0/543) 0.0	(8/642) 1.2	(627/642) 97.7	(0/86) 0.0	(0/543) 0.0	(15/642) 2.3
			Pseudomonas spp.	(298/337) 88.4	N/A	N/A	(4/337) 1.2	(313/337) 92.9	N/A	N/A	(24/337) 7.1
			**Aeromonas* spp. performance was established using internal development data.								
Ceftazidime	CAZ	caz02n	*Enterobacteriaceae*, *Pseudomonas*	94.1	2.1	0.4	0.8	97.0	7.4	0.8	1.6
Ceftazidime/Avibactam	CZA	cza02n	*Enterobacteriaceae*, *Pseudomonas aeruginosa*	94.5	0.0	0.5	0.0	98.9	0.0	1.2	0.0
Ceftolozane/Tazobactam	CT	ct01n	*Enterobacterales*, *Pseudomonas* spp.	(984/1031) 95.4	(1/61) 1.6	(5/970) 0.5	N/A	(1023/1031) 99.2	(2/61) 3.3	(6/970) 0.6	N/A

Antimicrobial	Antimicrobial Code	Antibiotic Version	Comment[1]	Essential Agreement				Category Agreement			
				% Error				% Error			
				%EA[2]	VME	ME	mE	%CA	VME	ME	mE
Ciprofloxacin	CIP	cip02n	*Acinetobacter* spp., *Pseudomonas* spp.	(270/293) 92.2	N/A	N/A	(0/293) 0.0	(289/293) 98.6	N/A	N/A	(4/293) 1.4
			Enterobacterales, *Aeromonas* spp.*, *P. multocida**, *Salmonella* spp.	(914/933) 98.0	(1/169) 0.6	(2/749) 0.3	(8/933) 0.9	(895/933) 95.9	(1/169) 0.6	(2/749) 0.3	(35/933) 3.8
			**Aeromonas* spp. and *Pasteurella* spp. performance was established using internal development data.								
Colistin	CS	cs02n	*Acinetobacter* spp., *Enterobacteriaceae*, *Pseudomonas aeruginosa*	(736/793) 92.8	(5/102) 4.9	(16/691) 2.3	N/A	(757/793) 95.5	(15/102) 14.7	(21/691) 3.0	N/A
Fosfomycin	FOS	fos02n	*Enterobacteriaceae*	93.1	24.4	3.8	-	92.3	34.1	4.7	-
Gentamicin	GM	gm02n	*Acinetobacter* Species, *Enterobacterales*	(984/1009) 97.5	(3/130) 2.3	(2/879) 0.2	N/A	(998/1009) 98.9	(4/130) 3.1	(7/879) 0.8	N/A
			Pseudomonas spp.*	(170/171) 99.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			*Breakpoints for *Pseudomonas* spp. are IE. An MIC only will be provided.								
Imipenem/Relebactam	IPR	ipr01n	*Enterobacterales* (except *Morganella* spp.), *Acinetobacter* spp., *Pseudomonas aeruginosa*	(600/631) 95.1	(1/96) 1.0	(2/535) 0.4	N/A	(622/631) 98.6	(5/96) 5.2	(4/535) 0.7	N/A
Meropenem	MEM	mem02n	❷	97.6	0.5	0.0	0.6	95.9	0.5	0.0	3.9
Meropenem/Vaborbactam	MEV	mev01n	*Enterobacteriaceae*, *Pseudomonas aeruginosa*	(512/526) 97.3	(0/45) 0.0	(1/481) 0.2	N/A	(523/526) 99.4	(1/45) 2.2	(2/481) 0.4	N/A
			Acinetobacter species*	(34/36) 94.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			*Breakpoints for *Acinetobacter* spp. are IE. An MIC only will be provided.								
Piperacillin/Tazobactam	TZP	tzp03n	*Enterobacteriaceae*, *Pseudomonas*	96.2	0.8	0.9	1.0	94.8	1.2	1.5	3.8
Tigecycline	TGC	tgc02n	*Enterobacteriaceae*	98.0	0.0	0.3	1.2	91.2	0.0	0.3	8.5

[1] Comment — Unless stated otherwise, performance is for *Enterobacteriaceae*, *Pseudomonas*, and *Acinetobacter*.

[2] Abbreviations — Bp = breakpoint committee; EA = essential agreement; CA = category agreement; VME = Very Major Error (susceptible result with resistant reference result); ME = Major Error (resistant result with susceptible reference result); mE = minor Error (susceptible or resistant result with an intermediate reference result, or an intermediate result with a susceptible or resistant reference result).

Key:

N/A = Not applicable

= EUCAST = European Committee on Antimicrobial Susceptibility

❶ ❷ = Symbol identifies the performance characteristics for a specific antimicrobial version.

List of Claims

Note: If the organism is not in the VITEK® 2 susceptibility database, results will not be reported.

Note: Organisms listed with an asterisk (*) indicate an AES claimed organism. No asterisk is displayed for a group; however, when an individual species (with an asterisk) is contained within a group, it is expertised.

Gram-Negative Organisms Claimed for AST-GN (keyID)

- *Achromobacter denitrificans*
- *Achromobacter xylosoxidans*
- *Acinetobacter baumannii**
- *Acinetobacter baumannii complex**
- *Acinetobacter calcoaceticus**
- *Acinetobacter haemolyticus*
- *Acinetobacter johnsonii*
- *Acinetobacter junii*
- *Acinetobacter lwoffii*
- *Acinetobacter pittii**
- *Acinetobacter* spp.
- *Aeromonas caviae*
- *Aeromonas hydrophila*
- *Aeromonas hydrophila/caviae*
- *Aeromonas sobria*
- *Alcaligenes faecalis* ssp. *faecalis*
- *Bordetella avium*
- *Bordetella bronchiseptica*
- *Brevundimonas diminuta*
- *Brevundimonas diminuta/vesicularis*
- *Brevundimonas vesicularis*
- *Burkholderia cepacia**
- *Cedecea davisae**
- *Cedecea lapagei**
- *Cedecea neteri**
- *Chryseobacterium gleum*
- *Chryseobacterium indologenes*
- *Citrobacter amalonaticus**
- *Citrobacter braakii**
- *Citrobacter farmeri**
- *Citrobacter freundii**
- *Citrobacter koseri**
- *Citrobacter youngae**
- *Comamonas testosteroni*
- *Cronobacter muytjensii**
- *Cronobacter sakazakii* group*
- *Cronobacter sakazakii**
- *Delftia acidovorans*
- *Edwardsiella hoshinae**
- *Edwardsiella tarda**
- *Elizabethkingia meningoseptica* (formerly known as *Chryseobacterium meningosepticum*)
- *Enterobacter asburiae**
- *Enterobacter cancerogenus**
- *Enterobacter cloacae**
- *Enterobacter cloacae* complex*
- *Enterobacter cloacae* ssp. *cloacae**
- *Enterobacter cloacae* ssp. *dissolvens* *
- *Escherichia coli**
- *Escherichia coli* ATCC® 25922™
- *Escherichia coli* ATCC® 35218™
- *Escherichia fergusonii*

- *Escherichia hermannii**
- *Escherichia vulneris**
- *Ewingella americana*
- *Hafnia alvei**
- *Hafnia paralvei**
- *Klebsiella (Enterobacter) aerogenes*
- *Klebsiella oxytoca**
- *Klebsiella pneumoniae**
- *Klebsiella pneumoniae* ssp. *ozaenae**
- *Klebsiella pneumoniae* ssp. *pneumoniae**
- *Klebsiella pneumoniae* ssp. *pneumoniae* ATCC® 700603 ™
- *Klebsiella pneumoniae* ssp. *rhinoscleromatis**
- *Klebsiella* spp.*
- *Kluyvera ascorbata**
- *Kluyvera cryocrescens**
- *Kluyvera intermedia* (formerly known as *Enterobacter intermedius*)*
- *Leclercia adecarboxylata**
- *Lelliottia amnigena* (formerly known as Enterobacter amnigenus)*
- *Lelliottia amnigena 1* (formerly known as Enterobacter amnigenus 1)*
- *Lelliottia amnigena 2* (formerly known as Enterobacter amnigenus 2)
- *Mannheimia haemolytica*
- *Moraxella* group
- *Moraxella lacunata*
- *Moraxella nonliquefaciens*
- *Moraxella osloensis*
- *Morganella morganii**
- *Morganella morganii* ssp. *morganii**
- *Morganella morganii* ssp. *sibonii**
- *Myroides* ssp.
- *Pantoea agglomerans*
- *Pantoea dispersa*
- *Pasteurella aerogenes*
- *Pasteurella multocida**
- *Pasteurella pneumotropica*
- *Plesiomonas shigelloides*
- *Pluralibacter gergoviae* (formerly known as *Enterobacter gergoviae*)*
- *Proteus hauseri**
- *Proteus mirabilis**
- *Proteus penneri**
- *Proteus vulgaris**
- *Providencia alcalifaciens**
- *Providencia rettgeri**
- *Providencia rustigianii**
- *Providencia stuartii**
- *Pseudomonas aeruginosa**
- *Pseudomonas aeruginosa* ATCC® 27853™
- *Pseudomonas alcaligenes*
- *Pseudomonas fluorescens*
- *Pseudomonas luteola*
- *Pseudomonas mendocina*
- *Pseudomonas oleovorans*

- *Pseudomonas oryzihabitans*
- *Pseudomonas putida*
- *Pseudomonas* spp.
- *Pseudomonas stutzeri*
- *Ralstonia pickettii*
- *Raoultella ornithinolytica**
- *Raoultella planticola**
- *Raoultella terrigena**
- *Salmonella enterica* ssp. *arizonae*
- *Salmonella enterica* ssp. *enterica**
- *Salmonella* group*
- *Salmonella* ser. Enteritidis*
- *Salmonella* ser. Paratyphi A*
- *Salmonella* ser. Paratyphi B*
- *Salmonella* ser. Paratyphi C*
- *Salmonella* ser. Typhi*
- *Salmonella* ser. Typhimurium*
- *Salmonella* spp.*
- *Serratia ficaria**
- *Serratia fonticola**
- *Serratia grimesii**
- *Serratia liquefaciens**
- *Serratia liquefaciens* group*
- *Serratia marcescens**
- *Serratia odorifera**
- *Serratia plymuthica**
- *Serratia proteamaculans*
- *Serratia rubidaea**
- *Shewanella putrefaciens*
- *Shewanella putrefaciens* group
- *Shigella boydii**
- *Shigella dysenteriae**
- *Shigella flexneri**
- *Shigella* group*
- *Shigella sonnei**
- *Shigella* spp.
- *Sphingobacterium multivorum*
- *Sphingobacterium spiritivorum*
- *Sphingomonas paucimobilis*
- *Stenotrophomonas maltophilia**
- *Vibrio alginolyticus*
- *Vibrio fluvialis*
- *Vibrio harveyi*
- *Vibrio metschnikovii*
- *Vibrio mimicus*
- *Vibrio parahaemolyticus*
- *Vibrio vulnificus*
- *Yersinia aldovae*
- *Yersinia enterocolitica**
- *Yersinia enterocolitica* group*
- *Yersinia frederiksenii**

- *Yersinia intermedia**
- *Yersinia kristensenii**
- *Yersinia pseudotuberculosis**
- *Yersinia ruckeri*

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User MUST enter the following barcodes into "Flex Panel Entry" program before first use of this Susceptibility Card.

01 A310S-0H--63

02 BAST-N43901R

03 C---Z372Z2U1

04 D2W322-333MJ

05 E1O353J2G3KK

06 F2P340000000F

If you have a compatible barcode reader for 2D barcodes (e.g., 1400g Honeywell 2D barcode reader), scan the following 2D barcode instead of the individual barcodes.

Index of Symbols

Symbol	Meaning
REF	Catalog number
IVD	In Vitro Diagnostic Medical Device
	Legal Manufacturer
	Temperature limitation
	Use by date
LOT	Batch code

Symbol	Meaning
	Consult Instructions for Use
	Date of manufacture
	Contains sufficient for <n> tests
	Authorized representative in the European Community
	For US Only : Caution : US Federal Law restricts this device to sale by or on the order of a licensed practitioner
	Importer

Instructions for use provided in the kit or downloadable from http://www.biomerieux.com.

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Revision History

Change type categories

N/A	Not applicable (First publication)
Correction	Correction of documentation anomalies
Technical change	Addition, revision and/or removal of information related to the product
Administrative	Implementation of non-technical changes noticeable to the user

Note: *Minor typographical, grammar, and formatting changes are not included in the revision history.*

Release Date	Part Number	Change Type	Change Summary
2022-08	061861-01	N/A	Not applicable (First publication)

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